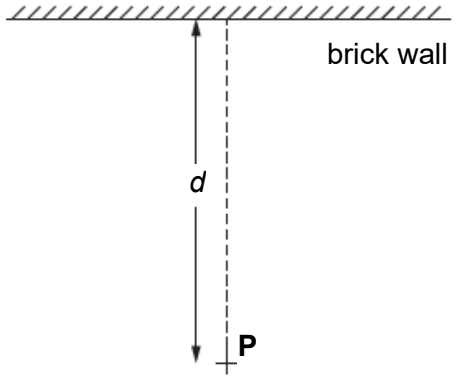
Answer **one** question from this Section in the spaces provided.

8 **(a)** To determine the speed of sound, a student facing a brick wall claps her hands at point **P**, which is a measured distance, *d*, from the brick wall, as shown in Fig. 8.1(a). A microphone at **P** is connected to a timing device, arranged so as to record the time, *t*, between the original clap and its echo. The experiment is carried out for three distances *d*, and the results are plotted as shown in Fig.8.1(b).



Det

Fig. 8.1(a)

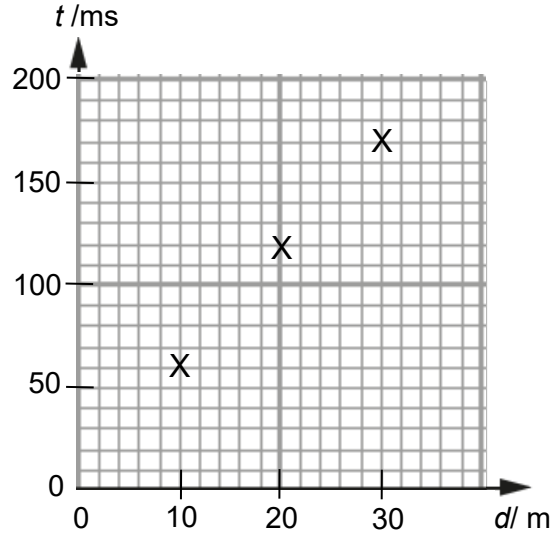
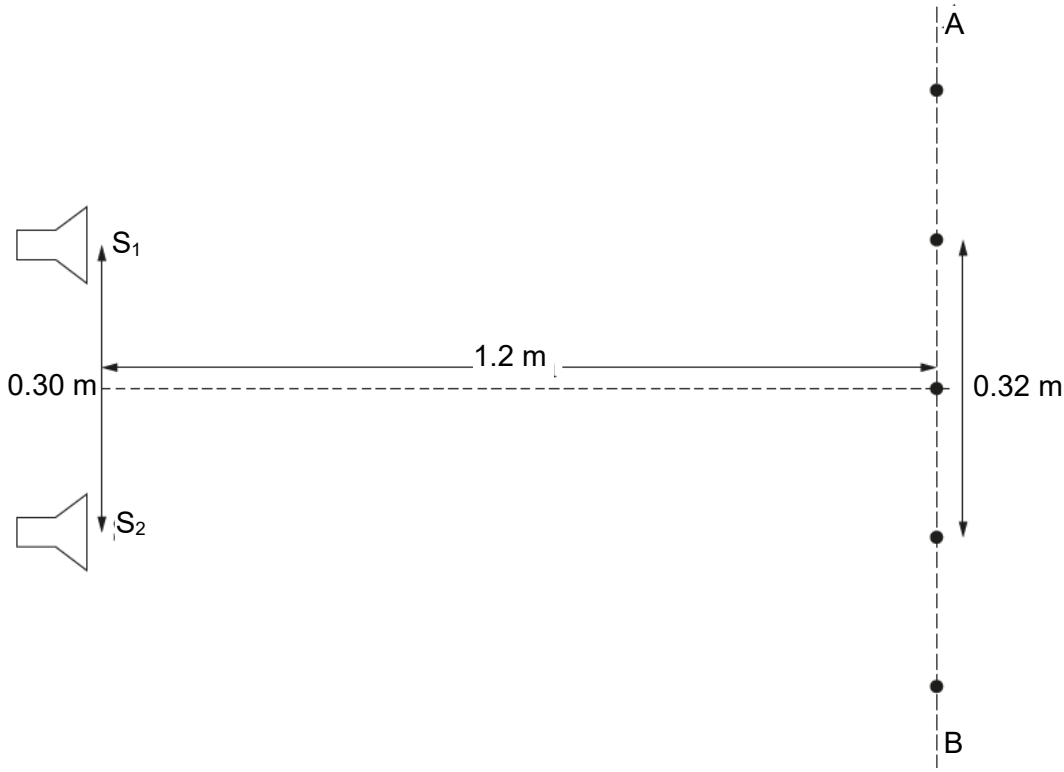


Fig. 8.1(b)

speed of sound = ... m s^{-1} [3]

(b) In another experiment, as shown in Fig. 8.2, the student sets up two small loudspeakers, S_1 and S_2, connected to the same signal generator, set to 8 300 Hz.

She moves a microphone along the line AB, and finds maxima of sound at the positions shown by dots along AB, with minima in between.



(i)

wavelength = ... m [2]

(ii) Hence, calculate the speed of sound from this experiment.

speed of sound = ... m s^{-1} [1]

(iii) **1.** Label with a letter 'M' on one of the dots along the line AB for which the path difference is $S_2M - S_1M = 2\lambda$. [1]

2. Explain why 'M' is a maxima.

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..... [1]

(iv) When the signal generator is set to 300 Hz, the student does not find a succession of maxima and minima as the microphone is moved along the line AB. Explain why this happens.

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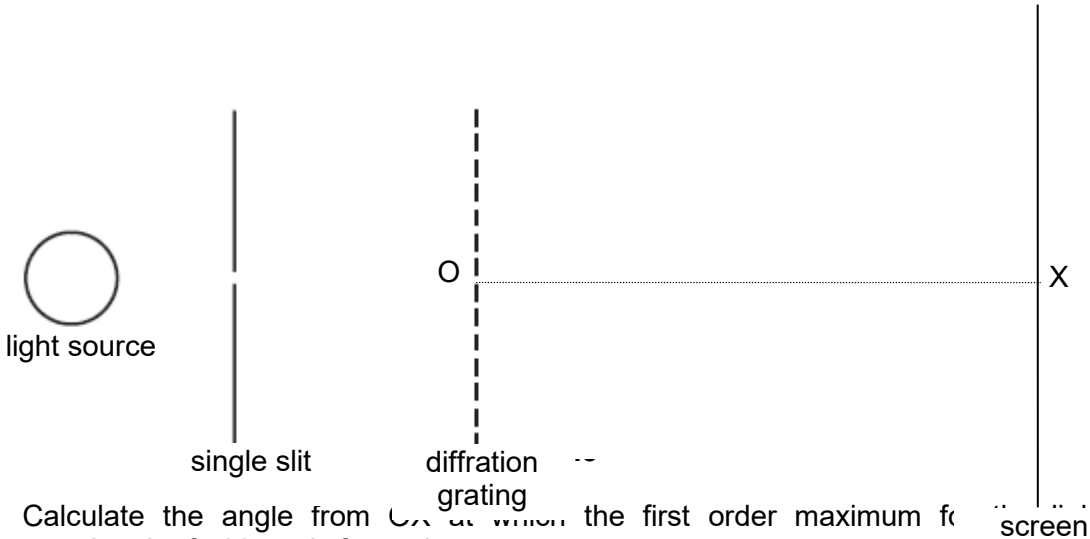
..... [3]

(c) Fig. 8.3 shows an arrangement used to analyse the light emitted by a source. The light source emits a range of wavelengths from 500 nm to 700 nm. The light is incident on a diffraction grating that has 10000 lines per metre. The diffracted pattern is formed on the screen placed at a distance away from the grating. OX is the line that indicates the direction of the undiffracted light.



(i) Calculate the angle from OX at which the first order maximum fo of wavelengh of 500 nm is formed.

angle from OX = ... ° [2]

(ii) Calculate the maximum angular separation of the first order spectrum on the same side of OX.

maximum angular separation = ... ° [2]

(iii) Calculate the maximum linear separation of the first order spectrum if the screen is placed at 2.0 m from the diffraction grating.

maximum linear separation = ... m

[2]

(iv) The single slit is initially illuminated by light from a point source that is 0.020 m from the slit.

State and explain how the intensity of light incident on the single slit changes when the light source is moved to a position 0.050 m from the slit.

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……..... [3]